From weighted coercivity to uniformly local Einstein repair

24 September 2026. This continuation removes the unnormalized global L^2 residual from the preceding diameter-free repair criterion, on a specified growth–coercivity regime. The additional information is a quantitative bound on propagation through the whole manifold, not another smallness condition on the residual's support.

1. Statement

For a smooth metric on a closed four-manifold use the Bianchi-gauged operator

$$\Phi_g(h) = \mathrm{Ric}_h + h + \delta_h^*(B_g h), \qquad B_g u = \delta_g u + \tfrac{1}{2} d\,\mathrm{tr}_g u.$$

Its derivative at g is

$$L_g = \tfrac{1}{2}\Delta_{L,g} + I = \tfrac{1}{2}\nabla^*\nabla + V_g, \tag{1}$$

where V_g is the symmetric curvature endomorphism. Unlike the earlier absolute spectral gap, the condition used here is **positive coercivity**:

$$\int_M \langle L_g u, u\rangle \, d\mu_g \geqslant \lambda \int_M |u|^2 \, d\mu_g \quad \text{for every smooth symmetric two-tensor } u. \tag{2}$$

No kernel or low mode has been projected out.

Theorem A (pointwise repair with controlled volume growth). Fix $v, K_1, C_G, \lambda > 0$, $d, H \geqslant 0$, and $0 < \alpha < 1$, with

$$H^2 < 8\lambda. \tag{3}$$

There are $\varepsilon, C > 0$, depending only on these data, such that the following holds. Suppose a closed connected oriented smooth four-manifold M, with $\pi_2(M) = 0$, carries a smooth metric g satisfying

$$\inf_x \mathrm{Vol}_g B_1(x) \geqslant v, \qquad \|\nabla \mathrm{Ric}_g\|_\infty \leqslant K_1, \tag{4}$$

the positive coercivity (2), and the actual-manifold growth bound

$$\mathrm{Vol}_g B_R(x) \leqslant C_G(1+R)^d e^{HR} \quad (x \in M,\ R \geqslant 0). \tag{5}$$

If

$$\delta_-(g) = \|g^{-1}\,\mathrm{Ric}_g + I\|_{\infty,\mathrm{op}} < \varepsilon, \tag{6}$$

then the same smooth manifold has a smooth metric h with

$$\mathrm{Ric}_h = -h, \qquad \|h - g\|_{C^{2,\alpha}_{\mathrm{uloc}}} \leqslant C\delta_-(g)^{1-\alpha}. \tag{7}$$

Here the uniformly local Hölder norm is the supremum of coefficient norms in a uniformly controlled small harmonic atlas, not a sum over that atlas. There is no diameter, total-volume, total-residual, or support-volume bound. The theorem does not require a gauge class or a prescribed complex structure.

The topological assumption is one supplier of local bounded geometry. If uniform $C^{2,\alpha}$ harmonic charts are provided directly, the same theorem holds without $\pi_2(M) = 0$ and without using (4) to obtain them; in that formulation the residual hypothesis is $\|\mathrm{Ric} + g\|_{C^{0,\alpha}_{\mathrm{uloc}}} < \varepsilon$. Thus the main analytic conclusion is a genuine uniformly local $C^{0,\alpha} \to C^{2,\alpha}$ inverse and repair estimate.

2. The inverse estimate: both decay and growth are retained

Choose once and for all

$$H/2 < a < \sqrt{2\lambda}, \qquad c_a = \lambda - a^2/2 > 0. \tag{8}$$

For a fixed centre x, put $\phi_x(y) = e^{-a d_g(x,y)}$. This positive Lipschitz function satisfies $|d\phi_x| \leqslant a\phi_x$ almost everywhere. No differentiability of the distance function at its cut locus is assumed.

For any smooth tensor u, integration by parts gives the exact identity

$$\int \langle L_g(\phi_x u), \phi_x u\rangle = \int \phi_x^2 \langle L_g u, u\rangle + \frac{1}{2}\int |d\phi_x|^2 |u|^2. \tag{9}$$

For Lipschitz ϕ_x, use density in H^1 in the quadratic-form identity. The zeroth-order terms agree on both sides; the factor 1/2 is the leading coefficient in (1).

If $L_g u = f$, (2), (8), Cauchy–Schwarz, and (9) give

$$c_a \|\phi_x u\|_2^2 \leqslant \int \phi_x^2 \langle f, u\rangle \leqslant \|\phi_x f\|_2 \|\phi_x u\|_2, \quad \|\phi_x u\|_2 \leqslant c_a^{-1}\|\phi_x f\|_2. \tag{10}$$

The right side can now be bounded by a **local supremum**, rather than the global L^2 norm. Splitting into unit annuli and using (5),

$$\int_M e^{-2a d_g(x,y)}\, d\mu_g(y) \leqslant C_G e^H \sum_{m=0}^{\infty} (m+2)^d e^{-(2a-H)m} \;=: J_a < \infty. \tag{11}$$

The constant is independent of x, diameter, and total volume. Therefore

$$\|u\|_{L^2(B_1(x))} \leqslant e^a c_a^{-1}\sqrt{J_a}\,\|f\|_\infty. \tag{12}$$

In the regime of Theorem A, $\delta_- < 1/2$ bounds Ricci uniformly. The local regularity theorem in the preceding continuation, using $\pi_2 = 0$ and the unit-ball lower bound, supplies uniform harmonic radii. The Ricci derivative bound and the harmonic metric equation then supply uniform $C^{2,\alpha}$ metric bounds on smaller charts. In particular the coefficients of (1), including V_g, have uniform $C^{0,\alpha}$ bounds.

Local Moser estimates for the connection Laplacian with bounded potential give, for a fixed $0 < \rho \leqslant 1$,

$$|u|(x) \leqslant C_{\mathrm{loc}}\Big(\|u\|_{L^2(B_\rho(x))} + \|f\|_{C^0(B_\rho(x))}\Big).$$

Combining this with (12), then applying local Schauder estimates, proves

$$\boxed{\|u\|_{C^{2,\alpha}_{\mathrm{uloc}}} \leqslant C_I \|L_g u\|_{C^{0,\alpha}_{\mathrm{uloc}}}.} \tag{13}$$

On each closed manifold, positive coercivity and elliptic Fredholm theory give existence and uniqueness of $u = L_g^{-1} f$. Equation (13) makes the inverse uniform across the family. We have not deduced (13) from an unweighted L^2 gap alone: (11) is the additional global step.

For polynomial growth, $H = 0$, every $\lambda > 0$ permits a choice in (8). For exponential growth, the sufficient condition (3) is the precise comparison needed by this argument. It is not claimed optimal. In particular failure of (3) does not imply failure of an Einstein inverse.

3. Nonlinear repair

In the uniformly local Hölder spaces put

$$E = \mathrm{Ric}_g + g, \qquad \Phi_g(g+u) = E + L_g u + N_g(u).$$

Uniform $C^{2,\alpha}$ bounds give, in a fixed sufficiently small ball,

$$\|N_g(u) - N_g(z)\|_{C^{0,\alpha}_{\mathrm{uloc}}} \leqslant C_N\Big(\|u\|_{C^{2,\alpha}_{\mathrm{uloc}}} + \|z\|_{C^{2,\alpha}_{\mathrm{uloc}}}\Big)\|u - z\|_{C^{2,\alpha}_{\mathrm{uloc}}}. \tag{14}$$

This is a local Hölder product estimate: neither summation over all charts nor a volume factor appears. By (4) and tensor interpolation,

$$\|E\|_{C^{0,\alpha}_{\mathrm{uloc}}} \leqslant C\delta_-^{1-\alpha}. \tag{15}$$

Equations (13)–(15) make $u \mapsto -L_g^{-1}(E + N_g(u))$ a contraction in the ball of radius $2C_I\|E\|_{C^{0,\alpha}_{\mathrm{uloc}}}$ when (6) is sufficiently small. The resulting $h = g + u$ is positive definite and has negative definite Ricci tensor. Applying B_h to $\Phi_g(h) = 0$ gives, for $b = B_g h$,

$$\nabla_h^*\nabla_h b - \mathrm{Ric}_h(b) = 0.$$

Integration on the closed manifold gives $b = 0$, so h is Einstein. Elliptic bootstrapping gives smoothness. This proves Theorem A.

4. A nonempty large-diameter family with distributed residual

Fix an oriented closed hyperbolic four-manifold (N, h) with $\mathrm{Ric}_h = -h$ and an epimorphism $\varphi : \pi_1(N) \to \mathbb{Z}$. Such compact examples exist by Millson's construction, which gives nonzero first Betti number in every dimension at least three. Let $(\widehat{N}, \widehat{h})$ be the cyclic infinite cover and (N_m, h_m) its quotient by $m\mathbb{Z}$.

Lemma B (the correct cover-growth bound). There is $C_N < \infty$, depending only on this fixed cyclic cover, such that

$$\mathrm{Vol}_{h_m} B_R(x) \leqslant C_N(1 + R) \quad (m \geqslant 1,\ R \geqslant 0). \tag{16}$$

Proof. The deck group $\mathbb{Z}$ acts properly and cocompactly on $\widehat{N}$. Choose a bounded fundamental region of finite volume. The orbit map of this action is a quasi-isometry: equivalently, only $O(1+R)$ translates of the region can meet a radius-R ball, with constants depending on the fixed region and deck generator. Counting their volumes proves the linear bound on $\widehat{N}$, uniformly in the centre. Every radius-R ball in N_m is the image of a radius-R ball in $\widehat{N}$, by lifting shortest paths from its centre. Its volume is at most the volume upstairs, proving (16). □

The cover used in this argument is **not** the universal hyperbolic cover, and (16) is not asserted for arbitrary finite covers. Since $\mathrm{Vol}(N_m) = m\,\mathrm{Vol}(N)$, (16) also shows $\mathrm{diam}(N_m) \to \infty$.

Choose $\eta > 0$ small and consider any smooth metrics g_m with

$$\|g_m - h_m\|_{C^3_{\mathrm{uloc}}(h_m)} \leqslant \eta. \tag{17}$$

The constants below are uniform in m. The metrics are uniformly bilipschitz to h_m, so (16) remains a polynomial bound with adjusted constant, and unit balls stay uniformly noncollapsed. They have $\pi_2 = 0$, bounded Ricci derivative, and sectional curvature uniformly close to $-1/3$.

For the exact h_m, Koiso's estimate gives

$$\int \langle L_{h_m} u, u \rangle \geqslant \tfrac{1}{3} \|u\|_2^2. \tag{18}$$

Its almost-Einstein version gives, for η sufficiently small, positive coercivity with $\lambda = 1/4$ for every g_m in (17). Thus $H = 0, d = 1, \lambda = 1/4, a = 1/2$ work simultaneously, with

$$c_a = \tfrac{1}{8}, \qquad J_a \leqslant C_N \sum_{j=0}^{\infty} (j+2) e^{-j} = C_N \frac{2 - e^{-1}}{(1 - e^{-1})^2}. \tag{19}$$

Theorem A applies to every metric in (17) whose pointwise Ricci residual is below its fixed threshold. The required threshold is independent of the number of cyclic cells, and the metrics need not be deck-invariant.

This family contains residuals which are small everywhere but not small in global L^2. For an explicit example, choose a ball in one fixed fundamental cell and a smooth bump β supported there and equal to one on a smaller ball. On its m disjoint translates choose arbitrary coefficients $c_j \in [1, 2]$, and put

$$f_m = t \sum_{j=1}^{m} c_j \beta_j, \qquad g_m = e^{2 f_m} h_m. \tag{20}$$

For small fixed $t > 0$, (17), all uniform constants and $\delta_-(g_m) \leqslant Ct$ hold. On each smaller ball the conformal factor is constant, so $\mathrm{Ric}_{g_m} + g_m = (e^{2tc_j} - 1) h_m$. Consequently

$$\| \mathrm{Ric}_{g_m} + g_m \|_{L^2(g_m)} \geqslant ct\sqrt{m}. \tag{21}$$

For sufficiently small fixed t, Theorem A repairs every g_m, although (21) diverges. This is a real gain over the earlier mixed-norm criterion, not merely a change from one notation for the residual to another. The underlying covers already admit hyperbolic metrics; this application establishes a **uniform repair neighbourhood for distributed perturbations**, not previously unknown Einstein topologies.

5. Scope relative to classical analysis and the Yau26 program

The weighted quadratic-form identity is classical; see Jäckel, Remark 2.6. His Lemma 2.4 supplies the almost-Einstein Koiso estimate used in (18), after scaling from $\mathrm{Ric} = -(n-1)g$ to $\mathrm{Ric} = -g$. The local Moser estimate is his Lemma 2.2. The new composition here uses the **actual quotient's slow volume growth** to convert weighted residual norms into uniformly local norms, and then uses the preceding topological regularity result as a source of controlled charts.

Jäckel's Theorem 1.1 permits collapse, under its negative-sectional and hyperbolic-thin-part hypotheses, but retains a small global L^2 residual. The present argument requires local noncollapse and (3),(5), but permits errors spread throughout arbitrarily large volume. Premoselli's repair analysis uses a different weighted inverse tailored to its gluing regions and source support. None of these results is an unrestricted replacement for the others. No global-priority claim is made for the weighted principle.

There are two important mathematical exclusions from the present theorem. First, the earlier assumption $\|L_g u\|_2 \geqslant \lambda \|u\|_2$ is not interchangeable with (2); negative eigenvalues invalidate the weighted coercivity step. Second, products of hyperbolic curves have genuine Teichmüller kernels,

so they do not satisfy (2). An obstruction equation cannot be bypassed by deleting those tensors. The growth comparison also concerns all balls on the actual manifold; a global entropy number without uniform finite-radius bounds does not by itself supply (11).

The scientific increment is therefore specific: the prior no-diameter criterion required $\delta^{1-\alpha} + \|E\|_2$ to be small; here a verified growth–coercivity condition makes **pointwise small Ricci error** sufficient in a family of unbounded diameter and volume. It remains a restricted sufficient answer to the original same-smooth-manifold existence question, not a dimension-only Yau26 threshold.

6. Reproduction scope

`check_uniformly_local_repair.py` verifies the factor in (9), the growth threshold and constants in (19), and the four-dimensional scaling in (21). These are finite algebraic checks. They do not certify the elliptic, covering-space or nonlinear arguments above.